

Chapter 6 The Analysis of Variance and Randomized Block Design

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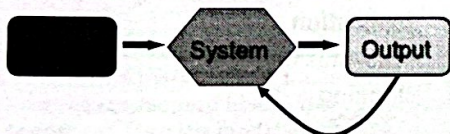
6.1 Fundamental Principles of the Analysis of Variance

The Analysis of Variance: a statistical method partitioning the total variation into several parts related to different sources of variation, so that we can understand the relative importance of each source in the total variation.

- ☐ Decomposition of degrees of freedom and sum of squares
- ☐ F test

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- ☐ Experimentation allows an investigator to find what happens to the output or response when the settings of the input variables in a system are purposely changed.



Control and Optimize the System

- ☐ Experimentation is used to understand and improve a system (product / process).

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☐ Single-factor Experiment

Simple, easy conducted, one level served as check
Variety trial, fertilize test, pesticide test

☐ Multiple-factor Experiment

More informative and efficient, interaction
Regional trial (Y, L, G, Y×L, Y×G, L×G, Y×L×G)

☐ Comprehensive Experiment

Unbalanced, superior treatment combination
Choose an optimized comprehensive high yield
technique suitable for local environment

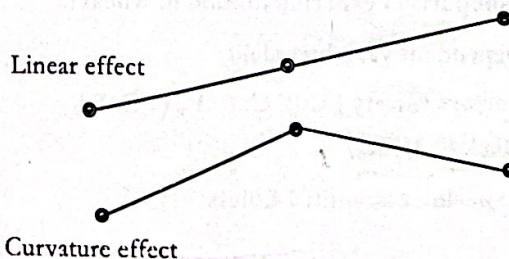
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数量性状 Quantitative Factors:

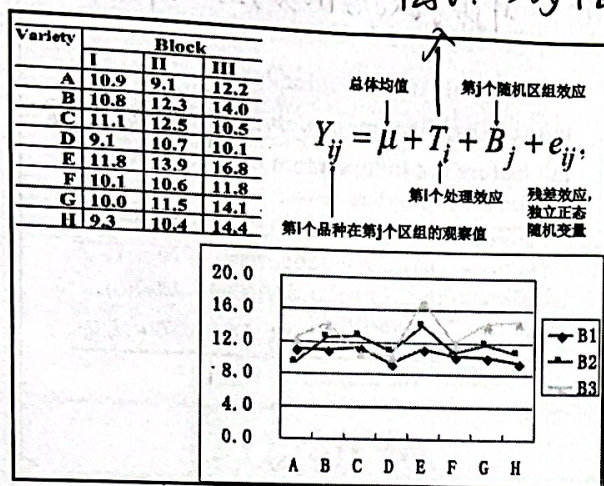
- ☐ Factor taking values over a continuous range.
- ☐ More freedom in choosing the levels.
- ☐ Example: for range 100°C-200°C, choosing 130°C and 160°C for two levels, or 125°C, 150°C, 175°C for three levels.
- ☐ Two levels are adopted for a linear effect, three or more levels for a curvature effect
- ☐ Enough differences among levels are needed, but not too large.

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Levels for factors



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☆

The linear model of the experimental design

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{21} \\ Y_{22} \\ Y_{23} \\ Y_{31} \\ Y_{32} \\ Y_{33} \\ Y_{41} \\ Y_{42} \\ Y_{43} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ T_1 \\ T_2 \\ T_3 \\ B_1 \\ B_2 \\ B_3 \end{bmatrix} + \begin{bmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \\ e_{31} \\ e_{32} \\ e_{33} \\ e_{41} \\ e_{42} \\ e_{43} \end{bmatrix}$$

$y = Xb + e$

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If the effects of treatments and randomized blocks are all fixed, then the dependent variable Y_{ij} is normally distributed

$$Y_{ij} \sim N(\mu + T_i + B_j, \sigma_e^2)$$

Expression in Matrix Form:

Suppose that there are four treatments ($i=1 \sim 4$) and three randomized blocks ($j=1, 2, 3$) in some experiment. Then the model in matrix form is

$$y = Xb + e \sim MVN_{12}(Xb, \sigma_e^2 I)$$

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Estimators of the Regression Parameters of the linear model

$$y = Xb + e \sim MVN(Xb, \sigma_e^2 I)$$

If $X^T X$ is a non-singular matrix, then the estimators of the regression coefficients are the only solution of the equations

$$\hat{b} = (X^T X)^{-1} X^T y = Ay$$

where

$$X^T X = \begin{bmatrix} n & \sum x_{1i} & \sum x_{2i} & \dots & \sum x_{pi} \\ \sum x_{1i} & \sum x_{1i}^2 & \sum x_{1i} x_{2i} & \dots & \sum x_{1i} x_{pi} \\ \sum x_{2i} & \sum x_{1i} x_{2i} & \sum x_{2i}^2 & \dots & \sum x_{2i} x_{pi} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum x_{pi} & \sum x_{1i} x_{pi} & \sum x_{2i} x_{pi} & \dots & \sum x_{pi}^2 \end{bmatrix} \quad X^T y = \begin{bmatrix} \sum y_i \\ \sum x_{1i} y_i \\ \sum x_{2i} y_i \\ \vdots \\ \sum x_{pi} y_i \end{bmatrix}$$

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$\hat{b}$ is a normally distributed random vector

$$\hat{b} \sim MVN_{p+1}(\hat{b}, \sigma^2 (X^T X)^{-1})$$

$$\begin{aligned} E(\hat{b}) &= E[(X^T X)^{-1} X^T y] = (X^T X)^{-1} X^T E(y) \\ &= (X^T X)^{-1} X^T Xb \\ &= \hat{b}
 \end{aligned}$$

$$\begin{aligned} Var(\hat{b}) &= Var[(X^T X)^{-1} X^T y] \\ &= (X^T X)^{-1} X^T Var(y) [(X^T X)^{-1} X^T]^T \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1}
 \end{aligned}$$

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The residual vector $\hat{e}$ is an unbiased estimation of the vector of random errors

$$\begin{aligned} E(\hat{e}) &= E(y - X\hat{b}) \\ &= Xb - Xb \\ &= 0
 \end{aligned}$$

The Residual Sum of Squares is the product of the transposed residual vector $\hat{e}^T$ and the residual vector $\hat{e}$

$$\begin{aligned} SSE &= \hat{e}^T \hat{e} \\ &= (y - \hat{y})^T (y - \hat{y}) \\ &= (y - X\hat{b})^T (y - X\hat{b}) \\ &= y^T y - \hat{b}^T X^T y
 \end{aligned}$$

SSTO

SSR(μ, T, B)

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平衡数据

For the randomized block design model (balance data), the regression sum of squares can be decomposed into several parts independent of each other

$$SSR(\mu, T, B) = SSR(\mu) + SSR(T) + SSR(B)$$

Partitioning the total sum of squares

$$SSTO = \sum_i \sum_j Y_{ij}^2 - SSR(\mu) = SSR(T) + SSR(B) + SSE(\mu, T, B)$$

Partitioning the total degrees of freedom

$$bt - 1 = (t - 1) + (b - 1) + (t - 1)(b - 1)$$

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平衡数据的方差分析表

ANOVA table of the randomized complete block design (balanced data) 平衡数据
(all factors are independent of each other)

Source	DF	SS	MS
Treatment	$t-1$	$SSR(T)$	$MSR(T)$
Block	$b-1$	$SSR(B)$	$MSR(B)$
Residual	$(t-1)(b-1)$	$SSE(\mu, T, B)$	$MSE(\mu, T, B)$

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Balanced and Unbalanced Data (非平衡数据)

Variety	Block		
	I	II	III
A	10.9	9.1	12.2
B	10.8	12.3	14.0
C	11.1	12.5	10.5
D	9.1	10.7	10.1
E	11.8	13.9	16.8
F	10.1	10.6	11.8
G	10.0	11.5	14.1
H	9.3	10.4	14.4

Variety	Block		
	I	II	III
A	10.9	9.1	12.2
B	10.8	12.3	14.0
C	11.1	12.5	.
D	9.1	.	.
E	11.8	13.9	16.8
F	10.1	10.6	11.8
G	10.0	11.5	14.1
H	9.3	10.4	14.4

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The regression sum of squares for the full model including population, treatment effects and block effects is:

$$SSR(\mu, T, B) = \mathbf{b}^T \mathbf{X}^T \mathbf{y} = (\mathbf{b}_0^T \quad \mathbf{b}_T^T \quad \mathbf{b}_B^T) \begin{pmatrix} \mathbf{X}_0^T \\ \mathbf{X}_T^T \\ \mathbf{X}_B^T \end{pmatrix} \mathbf{y} \\ = \mathbf{b}_0^T \mathbf{X}_0^T \mathbf{y} + \mathbf{b}_T^T \mathbf{X}_T^T \mathbf{y} + \mathbf{b}_B^T \mathbf{X}_B^T \mathbf{y}$$

So we can calculate the residual sum of squares

$$SSE(\mu, T, B) = \mathbf{y}^T \mathbf{y} - \mathbf{b}^T \mathbf{X}^T \mathbf{y}$$

The degrees of freedom of the total regression effect is $\rho(\mathbf{X})$, the degrees of freedom of the residual effect is $n - \rho(\mathbf{X})$

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Reduced Model 1

$$\mathbf{y} = \mathbf{X}_0 \mathbf{b}_{0(1)} + \mathbf{e} \\ SSTO = SSR(\mu) + SSE(\mu) \\ df_{TO} = 1 + (n - 1)$$

Reduced Model 2

$$\mathbf{y} = \mathbf{X}_0 \mathbf{b}_{0(2)} + \mathbf{X}_T \mathbf{b}_{T(2)} + \mathbf{e} \\ SSTO = SSR(\mu, T) + SSE(\mu, T) \\ = \underbrace{SSR(\mu)} + \underbrace{SSR(T | \mu)} + \underbrace{SSE(\mu, T)} \\ df_{TO} = 1 + [\rho(\mathbf{X}_0 : \mathbf{X}_T) - 1] + [n - \rho(\mathbf{X}_0 : \mathbf{X}_T)]$$

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系数矩阵的 rank.

Full Model

$$\mathbf{y} = \mathbf{X}_0 \mathbf{b}_{0(3)} + \mathbf{X}_T \mathbf{b}_{T(3)} + \mathbf{X}_B \mathbf{b}_{B(3)} + \mathbf{e} \\ SSTO = SSR(\mu, T, B) + SSE(\mu, T, B) \\ = \underbrace{SSR(\mu, T)} + \underbrace{SSR(B | \mu, T)} + \underbrace{SSE(\mu, T, B)} \\ = \underbrace{SSR(\mu, B)} + \underbrace{SSR(T | \mu, B)} + \underbrace{SSE(\mu, T, B)} \\ df_{TO} = \rho(\mathbf{X}_0 : \mathbf{X}_T) + [\rho(\mathbf{X}) - \rho(\mathbf{X}_0 : \mathbf{X}_T)] + [n - \rho(\mathbf{X})] \\ = \underbrace{\rho(\mathbf{X}_0 : \mathbf{X}_B)} + \underbrace{[\rho(\mathbf{X}) - \rho(\mathbf{X}_0 : \mathbf{X}_B)]} + \underbrace{[n - \rho(\mathbf{X})]}$$

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模型解新扩展. 针对不等组设计, Type I SS 根据主效应进场顺序的不同, 计算结果也不同.

Type I SS (I 类平方和): Sequential calculated extra regression sum of squares based on the analysis of reduced model.

ANOVA table for the randomized complete block experimental design (unbalanced data)

Source	Degrees of Freedom	Sum of Squares (Type I)	Mean of Squares (Type I)
Treatment	$\rho(X_0: X_T) - 1$	$SSR(T \mu)$	$MSR(T \mu)$
Block	$\rho(X) - \rho(X_0: X_T)$	$SSR(B \mu, T)$	$MSR(B \mu, T)$
Residual	$n - \rho(X)$	$SSE(\mu, T, B)$	$MSE(\mu, T, B)$

主效应依次加入.

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以完全分解组间平方和.

The extra effects of block ($B|\mu, T$) can directly be tested by F statistic

$$F^* = \frac{MSR(B|\mu, T)}{MSE(\mu, T, B)} \sim F(\rho(X) - \rho(X_0: X_T), n - \rho(X))$$

We can also perform F test for the extra effects of treatment ($T|\mu$)

If there's no block effect, we can test the significance of the extra effect of treatment ($T|\mu$) by F statistic

$$F^* = MSR(T|\mu) / MSE(\mu, T, B)$$

or

$$F^* = MSR(T|\mu) / MSE(\mu, T)$$

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But if the block effect exists, we can not use this F statistic to test the extra effect of treatment ($T|\mu$)

The null hypothesis H_0 : if there's no extra effect of treatment ($T|\mu$), then we can use the following F statistic

$$F^* = \frac{MSR(T|\mu)}{MSE(\mu, T)} \sim F(\rho(X_0: X_T) - 1, n - \rho(X_0: X_T))$$

where

$$MSE(\mu, T) = \frac{SSR(B|\mu, T) + SSE(\mu, T, B)}{n - \rho(X_0: X_T)}$$

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Type III SS (III类平方和): the extra regression sum of squares of each effect in the full model.

ANOVA table for the randomized complete block experimental design (unbalanced data)

Source	Degrees of Freedom	Sum of Squares (Type III)	Mean of Squares (Type III)
Treatment	$\rho(X) - \rho(X_0: X_B)$	$SSR(T \mu, B)$	$MSR(T \mu, B)$
Block	$\rho(X) - \rho(X_0: X_T)$	$SSR(B \mu, T)$	$MSR(B \mu, T)$
Residual	$n - \rho(X)$	$SSE(\mu, T, B)$	$MSE(\mu, T, B)$

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平衡数据 Type I SS = Type III SS.
非平衡数据 Type I SS $\neq$ Type III SS.

The null hypothesis H_0 : If there's no extra effect of treatment ($T|\mu, B$), then the F^* follow an F distribution

$$F^* = \frac{MSR(T|\mu, B)}{MSE(\mu, T, B)} \sim F(\rho(X) - \rho(X_0: X_B), n - \rho(X))$$

F^* value can be used to test whether the extra effect of treatment ($T|\mu, B$) is significant.

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ANOVA table of the complete randomized block design on wheat yield (balanced data)

Cause of Variation	Degrees of Freedom	Mean of Squares (Type I)	Mean of Squares (Type III)	F value (Type I, III)
Treatment	7	13.78	13.78	8.4
Block	2	4.87	4.87	2.97
Residual Error	14	1.64	1.64	

ANOVA table of the complete randomized block design on wheat yield (unbalanced data)

Cause of Variation	Degrees of Freedom	Mean of Squares (Type I)	Mean of Squares (Type III)	F value (Type I)	F value (Type III)
Treatment	7	4.54	3.96	5.28	4.6
Block	2	19.32	19.32	22.45	22.45
Residual Error	11	0.86	0.86		

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线性对比

If the treatment effects are significant, then we can perform further linear test on difference between these treatment effects. If $\sum_i c_i = 0$, then $\sum_i c_i T_i$ is an estimable function of treatment effect. It can be used to test the significance in difference between treatment effects.

$Y_{ij} = \mu + T_i + B_j + \varepsilon_{ij}$
 $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ 且相互独立, $i = 1, \dots, a; j = 1, \dots, b$

If $\sum_i c_i = 0$, then $\sum_i c_i Y_{ij} \sim N(\sum_i c_i T_i, \frac{1}{b} \sum_i c_i^2 \sigma_\varepsilon^2)$

Testing
 $H_0: \sum_i c_i T_i = 0 \quad H_1: \sum_i c_i T_i \neq 0$

We can perform linear contrast test for treatment effects based on these distributions

Treatment T_i	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
C_1	1	0	0	-1	0	0	0	0
C_2	1	1	1	1	-1	-1	-1	-1
C_3	1	0	1	0	0	-2	0	0

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```
data wheat;
input treat block y@@;
cards;
1 1 10.9 1 2 9.1 1 3 12.2
2 1 10.8 2 2 12.3 2 3 14
3 1 11.3 2 12.5 3 3 .
4 1 9.1 4 2 . 4 3 .
5 1 11.8 5 2 13.9 5 3 16.8
6 1 10.1 6 2 10.6 6 3 11.8
7 1 10.0 7 2 11.5 7 3 14.1
8 1 9.3 8 2 10.4 8 3 14.4
;

proc glm;
class treat block;
model y = treat block;
contrast 'treat' treat 1 0 1 0 0 -2 0 0;
run;
```

Contrast	DF	Contrast SS	Mean Square	F Value	Pr > F
Treat	1	1.58312588	1.58312588	1.84	0.2021

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6.2 Multiple Comparisons 多重比较

Multiple Comparison: comparisons between each pair of $(\tau_1, \tau_2, \dots, \tau_t)$

If there are t treatments $(\tau_1, \tau_2, \dots, \tau_t)$, n_i is the number of observations for treatment τ_i

N (or n_T) $= n_1 + n_2 + \dots + n_t$

Hypothesis
 $H_0: \tau_i = \tau_j, \quad H_1: \tau_i \neq \tau_j$

There are $t(t-1)/2$ comparisons

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Error Rate per Comparison (α_{PC}):

$\alpha_{PC} = \frac{\text{number of comparisons falsely declared significant}}{\text{number of comparisons}}$

Type I error

Error Rate of Experiment (α_{EW}):

$\alpha_{EW} = \frac{\text{number of experiments with at least one comparison falsely declared significant}}{\text{number of experiments}}$

Experimentwise (familywise) error rate

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I型错误: H_0 真但拒绝 H_0 .

(α)

For the test of $H_0: \tau_1 = \tau_2 = \dots = \tau_t = 0$ which has k pairs of comparisons

$1 - \alpha_{EW} = \Pr(\text{Accepting } H_0 | H_0 \text{ is true})$
 $= \Pr(\text{Accepting all } k \text{ comparisons} | H_0 \text{ is true})$
 $= (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_k)$
 $= (1 - \alpha_{PC})^k$
 $\approx 1 - k\alpha_{PC}$

α_i 是第 i 次比较的显著性水平, 对于 k 次检验, 各次检验一般取相同的显著性水平 α_{PC}

$\alpha_{EW} = k\alpha_{PC}$

If $k = 10$ and $\alpha_{PC} = 0.05$, then $\alpha_{EW} = 0.50$

If $k = 10$ and $\alpha_{EW} = 0.05$, then $\alpha_{PC} = 0.005$

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(1) Fisher's (protected) least-significant-difference (LSD) test (α_{PC})

For testing $H_0: \tau_1 = \tau_2 = \dots = \tau_t = 0$

If H_0 is rejected, then make all pairwise comparisons of means

$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \quad i = 1, \dots, t \quad j = 1, \dots, n_i$
 $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2), \varepsilon_{ij}$ 间相互独立 $N = \sum_{i=1}^t n_i$

$Y_i - Y_j \sim N(\tau_i - \tau_j, \sigma_\varepsilon^2 (\frac{1}{n_i} + \frac{1}{n_j}))$

$t^* = \frac{Y_i - Y_j}{\sqrt{MSE(\frac{1}{n_i} + \frac{1}{n_j})}} \sim t(N - t)$

$LSD = t(\frac{1}{2} N - t) \sqrt{MSE(\frac{1}{n_i} + \frac{1}{n_j})}$

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α_{PC} 在 0.05 以下,
 才能保证 α_{EW} 在 0.05 以下.

(1) Fisher's (protected) least-significant-difference test (α_{PC})

- The major disadvantage of the LSD technique is that its error rate is not satisfactory for testing all possible pairs of mean differences when there are a moderate or large number to be compared.
- Control the error rate (α_{PC}) for each individual pairwise comparison but doesn't control the experimentwise error rate (α_{EW})

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(2) Tukey's studentized range test on all main effect means (α_{EW})

Studentized range distribution:

Let $w = \max(\bar{Y}_i) - \min(\bar{Y}_j)$, $i(j, i \neq j) = 1, 2, \dots, t$ Then $\frac{w}{\sqrt{MSE/(\frac{1}{n_i} + \frac{1}{n_j})}} \sim q(t, N-t)$ □ 100(1- α)% confidence interval for all ($\tau_i - \tau_j$)

$$\bar{Y}_i - \bar{Y}_j \pm q_{\alpha}(t, N-t) \sqrt{MSE(\frac{1}{n_i} + \frac{1}{n_j})}$$

Control the experimentwise error rate: α_{EW}

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(3) Scheffé's multiple-comparison procedure on all main effect means (α_{EW})

Suppose that a set of m contrasts in the treatment means of interest have been determined

$$L_u = \sum_{i=1}^t c_{iu} \tau_i \quad \text{and} \quad \sum_{i=1}^t c_{iu} = 0 \quad u = 1, 2, \dots, m$$

The corresponding estimator of these contrasts are:

$$\hat{L}_u = \sum_{i=1}^t c_{iu} \bar{y}_i \quad u = 1, 2, \dots, m$$

Hypothesis: $H_0: L_u = 0 \quad H_1: L_u \neq 0$

$$S_{\alpha, u} = S_{L_u} \sqrt{(t-1)F(\alpha; t-1, N-t)} \quad S_{L_u} = \sqrt{MSE \sum_{i=1}^t \frac{c_{iu}^2}{n_i}}$$

if $|\hat{L}_u| > S_{\alpha, u}$, then, reject the H_0 , otherwise accept H_0

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(3) Scheffé's multiple-comparison procedure on all main effect means (α_{EW})

□ 100(1- α)% confidence interval for contrast L_u

$$\hat{L}_u - S_{\alpha, u} \sqrt{MSE \sum_{i=1}^t \frac{c_{iu}^2}{n_i}} < L_u < \hat{L}_u + S_{\alpha, u} \sqrt{MSE \sum_{i=1}^t \frac{c_{iu}^2}{n_i}}$$

$$\text{where } S_{\alpha, u} = \sqrt{(t-1)F(\alpha; t-1, n_T-t)}$$

Simultaneously confidence intervals in that the probability is at least 1- α that all of them are simultaneously true.

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(4) Bonferroni t tests of differences between means for all main effect means (α_{EW})

□ If there are k comparisons for $L_u = \sum_{i=1}^t c_{iu} \tau_i$ ($u = 1, \dots, k$)□ 100(1- α)% confidence interval on L_u

$$\hat{L}_u \pm t(\frac{\alpha}{2k}, N-t) \sqrt{MSE \sum_{i=1}^t \frac{c_{iu}^2}{n_i}}$$

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The following is a decision tree for selecting a multiple contrast method:

> pairwise

- equal groups sizes: Tukey
- unequal group sizes: Tukey-Kramer (修正的 Tukey法, 比较的两组大小不同时采用) or Scheffé

> not pairwise

- planned: Bonferroni
- not planned: Scheffé

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比较 conservative


```

Data Food;
Input Design Sale @@;
Datalines;
1 12 3 19
1 18 3 17
2 14 3 21
2 12 4 24
2 13 4 30
;

PROC GLM;
class Design;
MODEL Sale = Design;
Means Design/LSD;
Means Design/Tukey;
Means Design/SCHEFFE;
Means Design/BON;
RUN;
    
```

ANOVA 分析

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ANOVA model 无参数.

I	J	D	CI by LSD	CI by Tukey	CI by Shef	CI by Bon
4	-3	8	1.8	14.2 ***	-1.5	17.5
4	-1	12	5.2	18.8 ***	1.5	22.5***
4	-2	14	7.8	20.2 ***	4.5	23.5***
3	-4	-8	-14.2	-1.8 ***	-16.8	0.8
3	-1	4	-2.2	10.2	-4.8	12.8
3	-2	6	0.5	11.5 ***	-1.8	13.8
1	-4	-12	-18.8	-5.2 ***	-21.6	-2.4 ***
1	-3	-4	-10.2	2.2	-12.8	4.8
1	-2	2	-4.2	8.2	-6.8	10.8
2	-4	-14	-20.2	-7.8 ***	-22.8	-5.3 ***
2	-3	-6	-11.5	-0.5 ***	-13.8	1.8
2	-1	-2	-8.2	4.2	-10.8	6.8

4x4 数据表
假阳性高.

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6.3 Fixed Model and Random Model

Fixed Model (固定模型):
All parameters except residual in model are defined as fixed effects.

Random Model (随机模型): → residual
All the parameters except the intercept (population mean) of the model are random effects.

The statistical inference approaches depend on the analytic model.

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ANOVA model 无参数.

For the fixed model in randomized complete block design

$$Y_{ij} = \mu + T_i + B_j + e_{ij}, \quad i = 1, 2, \dots, t \text{ treatment}$$

$$j = 1, 2, \dots, b \text{ blocks}$$

Only the residual e_{ij} is a random effect following normal distribution, so the observed value (Y_{ij}) of a character also follows a normal distribution:

$$Y_{ij} \sim N(\mu + T_i + B_j, \sigma_e^2)$$

fixed model

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For the random model in randomized complete block design

$$Y_{ij} = \mu + T_i + B_j + e_{ij}, \quad i = 1, 2, \dots, t$$

$$j = 1, 2, \dots, b$$

Treatment effect T_i , Block effect B_j and residual effect e_{ij} follow independent normal distribution, thus,

$$Y_{ij} \sim N(\mu, \sigma_T^2 + \sigma_B^2 + \sigma_e^2)$$

random model

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Expectation of mean-square table for RCBD

Source	DF	MS	EMS (Fixed Model)	EMS (Random Model)
Treatment	$t-1$	MST	$\sigma_e^2 + \frac{b}{t-1} \sum T_i^2$	$\sigma_e^2 + b\sigma_T^2$
Block	$b-1$	MSB	$\sigma_e^2 + \frac{t}{b-1} \sum B_j^2$	$\sigma_e^2 + t\sigma_B^2$
Residual	$(t-1)(b-1)$	MSE	σ_e^2	σ_e^2

random model

所有EMS都有 residual
MST: 处理间差异

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Fixed model

Null Hypothesis $H_0: \sum_i T_i^2 = 0$

Alternative Hypothesis $H_1: \sum_i T_i^2 \neq 0$ 效应间有差异

Statistic $F^* = MST/MSE$

Null Hypothesis $H_0: \sum_j B_j^2 = 0$

Alternative Hypothesis $H_1: \sum_j B_j^2 \neq 0$

Statistic $F^* = MSB/MSE$

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Random model

Null Hypothesis $H_0: \sigma_\tau^2 = 0$

Alternative Hypothesis $H_1: \sigma_\tau^2 > 0$

Statistic $F^* = MST/MSE$

Null Hypothesis $H_0: \sigma_B^2 = 0$

Alternative Hypothesis $H_1: \sigma_B^2 > 0$

Statistic $F^* = MSB/MSE$

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完全随机设计

6.4 Randomized Block Design

□ 研究化学药品的反应速率，每天只能分析5个化学药品。由于天与天之间的差异会影响反应速率，将天设置为区组。每天按照随机顺序检测5个药品

Block	Treatments				
Day one	A1	A4	A3	A5	A2
Day two	A4	A1	A5	A2	A3
Day three	A2	A5	A1	A3	A4

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Advantages of randomized complete block design

1. It can, with effective grouping, provide substantially more precise results than a completely randomized design of comparable size
2. It can accommodate any number of treatments and replications
3. Different treatments need not have equal sample sizes
4. The statistical analysis is relatively simple.
5. If an entire treatment or a block needs to be dropped from the analysis for some reason, such as spoiled results, the analysis is not complicated thereby
6. Variability in experimental units can be deliberately introduced to widen the range of validity of the experimental results without sacrificing the precision of the results.

所有随机效应
实验单元，除
设天与天之间
无系统误差

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少了数据的话就必须造一个数，这个数相当于没有 random 的。

Disadvantages of randomized complete block design

1. Missing observations within a block require more complex calculations.
2. The degrees of freedom for experimental error are not as large as with a completely randomized design. block 也会自由度。
3. More assumptions are required for the model than for a completely randomized design model, e.g., no interactions between treatments and blocks, constant variance from block to block

假设：treatment 和 block 是独立的。

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Normal distribution

$$Y_{ij} = \mu + \tau_i + \rho_j + \varepsilon_{ij} \sim N(\mu + \tau_i + \rho_j, \sigma_\tau^2 + \sigma_\rho^2 + \sigma_\varepsilon^2)$$

$$i = 1, \dots, t; j = 1, \dots, r$$

$$\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2), \rho_j \sim N(0, \sigma_\rho^2), \sum_{i=1}^t \tau_i = 0$$

$$E(Y_{ij}) = E(\mu + \tau_i + \rho_j + \varepsilon_{ij}) = \mu + \tau_i + \rho_j$$

$$Var(Y_{ij}) = Var(\mu + \tau_i + \rho_j + \varepsilon_{ij}) = \sigma_\tau^2 + \sigma_\rho^2 + \sigma_\varepsilon^2$$

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Partition of sum of squares and degrees of freedom

$$\sum_{i=1}^t \sum_{j=1}^r (y_{ij} - \bar{y})^2 = \sum_{i=1}^t \sum_{j=1}^r (\bar{y}_i - \bar{y})^2 + \sum_{i=1}^t \sum_{j=1}^r (\bar{y}_j - \bar{y})^2 + \sum_{i=1}^t \sum_{j=1}^r (y_{ij} - \bar{y}_i - \bar{y}_j + \bar{y})^2$$

$$= r \sum_{i=1}^t (\bar{y}_i - \bar{y})^2 + t \sum_{j=1}^r (\bar{y}_j - \bar{y})^2 + \sum_{i=1}^t \sum_{j=1}^r (y_{ij} - \bar{y}_i - \bar{y}_j + \bar{y})^2$$

$$SSTO = SST + SSB + SSE$$

$$df_{TO} = df_T + df_B + df_E$$

$$rt-1 = (t-1) + (r-1) + (t-1)(r-1)$$

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ANOVA table for randomized block design

Source	df	SS	MS	E(MS) Fixed
Treatments	$t-1$	SST	$MST=SST/(t-1)$	$\sigma_e^2 + \frac{r}{t-1} \sum \tau_i^2$
Block	$r-1$	SSB	$MSB=SSB/(r-1)$	$\sigma_e^2 + \sigma_\tau^2$
Error treatments	$(t-1)(r-1)$	SSE	$MSE=SSE/(t-1)(r-1)$	σ_e^2
Total	$nr-1$	$SSTO$		

Null $H_0: \tau_1 = \tau_2 = \dots = \tau_t = 0$ Alternative H_1 : Not all $\tau_i = 0$ 不全为0. $F^* = MST/MSE \sim F(t-1, (t-1)(r-1))$ if H_0 is true

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Case Study (1):

Study on the genetic variation of yield components for 50 cotton materials

Dependent variables: Three component traits of cotton yield (Boll number, Boll weight, Lint)

Factors: Genotype, Block

Experiment unit (plot): 150

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Data Cot;
Input Geno Blk BollsR BsizeR LintR;
Datalines;
1 1 15.051 4.512 35.499
1 2 10.58 4.145 34.714
1 3 13.123 4.713 39.167
2 1 16.084 4.635 38.979
2 2 12.647 4.369 39.33
2 3 15.69 4.909 39.748
50 1 9.874 4.985 28.278
50 2 12.355 5.265 31.189
50 3 12.089 5.356 32.998

PROC GLM;
Class Geno Blk;
MODEL BollsR BsizeR LintR = Geno Blk;
Means Geno/LSD;
Run;
PROC GLM;
Class Geno;
MODEL BollsR BsizeR LintR = Geno;
Means Geno/LSD;
Run;

对不同品种间的性状均值差异进行多重比较

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ANOVA table of yield traits in randomized complete block design

Source	df	Bolls MS	P-value	Bsize MS	P-value	Lint MS	P-value
Genotype	49	12.195	<0.0001	0.265	<0.0001	3.120	<0.0001
Block	2	3.598	0.363	0.289	0.024	17.975	0.002
Error	98	3.512		0.0745		3.120	

ANOVA table of yield traits in complete randomized design

Source	df	Bolls MS	P-value	Bsize MS	P-value	Lint MS	P-value
Genotype	49	12.195	<0.0001	0.265	<0.0001	3.120	<0.0001
Error	100	3.514		0.0792		3.473	

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Results of multiple comparisons

当2个均值对应的标记字母没有相同的字母时,表示两者存在显著性差异	B		A			20.017	3	15		
	B		A			19.796	3	14		
	B		A		C	18.617	3	10		
	B	D	A		C	18.022	3	41		
	B	D	A		C	17.920	3	47		
	B	D	A		C	17.919	3	40		
	B	D	A		C	17.899	3	38		
	E	B	D		A	C	17.827	3	28	
	E	B	D		A	C	F	17.644	3	4
M	L	I	K	J	H	G	F	14.788	3	18
M	L	I	K	J	H	G	F	14.710	3	36
M	L	I	K	J	H	G		14.551	3	25
M	L	I	K	J	H	G	N	14.453	3	8
M	L	I	K	J	H	G	N	14.447	3	48
M	L	I	K	J	H	G	N	14.439	3	30

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Block 能 控制 系统误差

控制 随机误差

有多次重复的 Latin Squares Design.

6.5 Latin Squares Design

Treatment: A, B, C

Block (1): 1, 2, 3

Block (2): Mon, Tue, Wed

	Operator		
Day	1	2	3
Mon	B	A	C
Tue	A	C	B
Wed	C	B	A

- Operator 1 would run treatment B on Monday, treatment A on Tuesday, and treatment C on Wednesday, and so on for the other operators.
- Note that each operator runs each treatment, and that all treatments are run on each day

Features

- There are r treatments
- There are two blocking variables, each containing r classes
- Each row and each column in the design square contains all treatments

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Advantages of a Latin square design

- Permit greater reductions in the variability of experimental errors 系统误差更小 (两个性质)
- Treatment effects can be studied from a small-scale experiment. This is particularly helpful in preliminary or pilot studies 预实验. (如果人和天都没显著差异 可以不用 block, 省自由度)
- It is often helpful in repeated measures experiments to take into account the order effect of treatments by means of a Latin square design

	Operator		
Day	1	2	3
Mon	B	A	C
Tue	A	C	B
Wed	C	B	A

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Disadvantages of a Latin square design

- The number of classes of each blocking variable must equal the number of treatments.
 - This leads to a very small number of degrees of freedom for experimental error when only a few treatments are studied.
 - On the other hand, when many treatments are studied, the degrees of freedom for experimental error may be larger than necessary.
- The assumptions of the model are restrictive (e.g., that there are no interactions between either blocking variable and treatments, and also none between the two blocking variables). 都没工作
- The two blocking variables cannot have different numbers of classes.
- The randomization required is somewhat more complex than that for earlier designs considered.

	Operator		
Day	1	2	3
Mon	B	A	C
Tue	A	C	B
Wed	C	B	A

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3x3			4x4				5x5				6x6				7x7				8x8				9x9										
A	B	C	A	B	C	D	A	B	C	D	E	A	B	C	D	E	F	A	B	C	D	E	F	G	A	B	C	D	E	F	G	H	I
B	C	A	B	C	D	A	B	C	D	E	F	B	C	D	E	F	G	B	C	D	E	F	G	H	B	C	D	E	F	G	H	I	
C	A	B	C	D	E	F	C	D	E	F	G	C	D	E	F	G	H	C	D	E	F	G	H	I	C	D	E	F	G	H	I	A	
D	C	A	D	E	F	G	D	E	F	G	H	D	E	F	G	H	I	D	E	F	G	H	I	A	D	E	F	G	H	I	A	B	
E	D	B	E	F	G	H	E	F	G	H	I	E	F	G	H	I	A	E	F	G	H	I	A	B	E	F	G	H	I	A	B	C	
F	E	C	F	G	H	I	F	G	H	I	A	F	G	H	I	A	B	F	G	H	I	A	B	C	F	G	H	I	A	B	C	D	
G	F	D	G	H	I	A	G	H	I	A	B	G	H	I	A	B	C	G	H	I	A	B	C	D	G	H	I	A	B	C	D	E	
H	G	E	H	I	A	B	H	I	A	B	C	H	I	A	B	C	D	H	I	A	B	C	D	E	H	I	A	B	C	D	E	F	
I	H	F	I	A	B	C	I	A	B	C	D	I	A	B	C	D	E	I	A	B	C	D	E	F	I	A	B	C	D	E	F	G	

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先选择标准方

A	B	C	D	E
B	A	E	C	D
C	D	A	E	B
D	E	B	A	C
E	C	D	B	A

随机调整列

A	D	E	C	B
B	C	D	E	A
C	E	B	A	D
D	A	C	B	E
E	B	A	D	C

随机安排处理

3	5	2	1	4
2	1	3	4	5
5	4	1	3	2
1	2	4	5	3
4	3	5	2	1

随机调整行

E	B	A	D	C
A	D	E	C	B
B	C	D	E	A
D	A	C	B	E
C	E	B	A	D

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Randomization by Excel

A	B	C	D	E	随机 调 整 列	C	B	E	D	A	0.24
B	A	E	C	D		E	A	D	C	B	-0.23
C	D	A	E	B		A	D	B	E	C	-1.85
D	E	B	A	C		B	E	C	A	D	-0.40
E	C	D	B	A		D	C	A	B	E	1.34
1.11	-1.20	-1.56	0.71	0.64	→	随机调整行					↓
4	1	5	3	2	← 随机 调 整 处 理	A	D	B	E	C	
5	3	2	4	1		B	E	C	A	D	
3	4	1	2	5		E	A	D	C	B	
2	5	3	1	4		C	B	E	D	A	
1	2	4	5	3		D	C	A	B	E	
						1	D				
						2	C				
						3	E				
						4	A				
						5	B				
								1	A	-0.30	
								2	B	1.73	
								3	C	-0.69	
								4	D	-2.12	
								5	E	-0.33	

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Case Studied (2)

- An experiment about the effects of five different types of background music (A B C D E) on the productivity of bank tellers.
- A given type of music is played for one day and the productivity is observed.
- Experimental units: A working day for the crew of bank tellers.
- Treatments or Latin letter: Combinations of tempo of music (slow, medium, fast) and style of music (instrumental and vocal, instrumental only).

Treatment	Latin Letter Designation	Tempo and Style of Music
1	A	Slow, instrumental and vocal
2	B	Medium, instrumental and vocal
3	C	Fast, instrumental and vocal
4	D	Medium, instrumental only
5	E	Fast, instrumental only

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TABLE 31.2 Latin square design background music study—experiment results (productivity of crew—data coded)

Week	Day					Total
	M	T	W	Th	F	
1	18 (D)	17 (C)	14 (A)	21 (B)	17 (E)	$Y_{1..} = 87$
2	13 (C)	34 (B)	21 (E)	16 (A)	15 (D)	$Y_{2..} = 99$
3	7 (A)	29 (D)	32 (B)	27 (E)	13 (C)	$Y_{3..} = 108$
4	17 (E)	13 (A)	24 (C)	31 (D)	23 (B)	$Y_{4..} = 110$
5	21 (B)	26 (E)	26 (D)	31 (C)	7 (A)	$Y_{5..} = 111$
Total	$Y_{.1.} = 76$	$Y_{.2.} = 119$	$Y_{.3.} = 117$	$Y_{.4.} = 126$	$Y_{.5.} = 77$	$Y_{...} = 515$

根据设计及变异来源建立分析模型

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijk}$$

$$\sim N(\mu + \tau_k, \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2)$$

Mix model

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$$Y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijk}$$

$$\sim N(\mu + \tau_k, \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2)$$

$$\alpha_i \sim N(0, \sigma_\alpha^2), \beta_j \sim N(0, \sigma_\beta^2),$$

$$\varepsilon_{ijk} \sim N(0, \sigma_\varepsilon^2), \sum_{k=1}^r \tau_k = 0$$

$$E(Y_{ijk}) = E(\mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijk}) = \mu + \tau_k$$

$$Var(Y_{ijk}) = Var(\mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijk})$$

$$= \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2$$

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Partition of Total Sum of Squares and Degrees of Freedom

$$Y_{ijk} - \bar{Y}_{...} = (\bar{Y}_{i..} - \bar{Y}_{...}) + (\bar{Y}_{.j.} - \bar{Y}_{...}) + (\bar{Y}_{..k} - \bar{Y}_{...})$$

$$+ (Y_{ijk} - \bar{Y}_{i..} - \bar{Y}_{.j.} - \bar{Y}_{..k} + 2\bar{Y}_{...})$$

$$\sum_i \sum_j (Y_{ijk} - \bar{Y}_{...})^2 = r \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2 + r \sum_j (\bar{Y}_{.j.} - \bar{Y}_{...})^2$$

$$+ r \sum_k (\bar{Y}_{..k} - \bar{Y}_{...})^2 + \sum_i \sum_j (Y_{ijk} - \bar{Y}_{i..} - \bar{Y}_{.j.} - \bar{Y}_{..k} + 2\bar{Y}_{...})^2$$

$$SSTO = SSR + SSC + SST + SSE$$

$$df_{TO} = df_R + df_C + df_T + df_e$$

$$r^2 - 1 = (r-1) + (r-1) + (r-1) + (r-1)(r-2)$$

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没有 replication

ANOVA table for Latin Squares Design

Source	df	MS	E(MS) Mixed
Rows	$r-1$	MSB	$\sigma_\alpha^2 + r\sigma_\varepsilon^2$
Columns	$r-1$	MSB	$\sigma_\beta^2 + r\sigma_\varepsilon^2$
Treatments	$r-1$	MST	$\sigma_\tau^2 + \frac{r}{r-1} \sum_i \tau_i^2$
Error	$(r-1)(r-2)$	MSE	σ_ε^2

原假设 $H_0: \tau_1 = \tau_2 = \dots = \tau_r = 0$ 备择假设 $H_1: \text{not all } \tau_k = 0$ $F^* = MST/MSE \sim F(r-1, (r-1)(r-2))$ 如果 H_0 是真

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Data Music;

Input Week Day \$ Treat \$ Prod @@;

Datalines;

```
1 M D 18 1 T C 17 1 W A 14 1 H B 21 1 F E 17
2 M C 13 2 T B 34 2 W E 21 2 H A 16 2 F D 15
3 M A 7 3 T D 29 3 W B 32 3 H E 27 3 F C 13
4 M E 17 4 T A 13 4 W C 24 4 H D 31 4 F B 25
5 M B 21 5 T E 26 5 W D 26 5 H C 31 5 F A 7
```

PROC GLM;

Class Week Day Treat;

MODEL Prod = Week Day Treat;

Means Treat/LSD;

Means Treat/Tukey;

Means Treat/SCHIFFE;

Means Treat/BON;

RUN;

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Source	DF	Type I SS	Mean Square	F Value	Pr > F
Week	4	82.00	20.50	1.31	0.3226
Day	4	477.20	119.30	7.60	0.0027
Treat	4	664.40	166.10	10.58	0.0007

Grouping	Tukey Grouping	Scheffe Grouping	Scheffe Grouping	Mean	N	Treat
A	A	A	A	26.500	5	B
B	A	A	A	23.800	5	D
B	A	A	A	21.500	5	E
B	A	A	A	19.500	5	C
B	C	B	B	11.400	5	A

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Efficiency of Blocking Variables							
CRD		RBD(W)		RBD(D)		LSD	
Source	df	MS	df	MS	df	MS	df
Model	4	166.10	8	93.30	8	142.70	12
Error	20	37.38	16	41.60	16	16.90	12

Week放入模型	Day放入模型	Week、Day放入模型
----------	---------	--------------

$$E_1 = \sigma_R^2 / \sigma_L^2 \quad \hat{E}_1 = 37.38 / 15.70 = 2.38$$

$$E_2 = \sigma_{B(W)}^2 / \sigma_L^2 \quad \hat{E}_2 = 41.60 / 15.70 = 2.65$$

$$E_3 = \sigma_{B(D)}^2 / \sigma_L^2 \quad \hat{E}_3 = 16.90 / 15.70 = 1.08$$

MSE 最小

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Additional Replications with Latin Square Design

(a) Data

IQ i	Age (j)		
	Young	Middle	Old
High	(B) 19	(A) 20	(C) 25
	16	24	21
Normal	(C) 24	(B) 14	(A) 14
	22	15	14
Low	(A) 10	(C) 12	(B) 7
	14	13	4

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$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijkl}$$

$$\sim N(\mu + \tau_k, \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2)$$

ANOVA table 有 replication

Source	df	MS	E(MS) Mixed
Rows	r-1	MSR	$\sigma_\alpha^2 + nr\sigma_\varepsilon^2$
Columns	r-1	MSC	$\sigma_\beta^2 + nr\sigma_\varepsilon^2$
Treatments	r-1	MST	$\sigma_\tau^2 + \frac{nr}{r-1} \sum \tau_i^2$
Error	nr ² -3r+2	MSE	σ_ε^2

n is the number of replication

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```

Data Test;
Input IQ $ AGE $ TREAT $ SCORES @@;
datalines;
HYB 19 HMA 20 HOC 25
HYB 16 HMA 24 HOC 21
NYC 24 NMB 14 NOA 14
NYC 22 NMB 15 NOA 14
LYA 10 L MC 12 L OB 7
LYA 14 L MC 13 L OB 4
;
PROC GLM;
class IQ AGE TREAT;
MODEL SCORES = IQ AGE TREAT;
Means Treat/LSD;
Means Treat/Tukey;
Means Treat/SCHIFFE;
Means Treat/BON;
RUN;
    
```

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Comparison of ANOVA results (No replication)

```

Data Test;
Input IQ $ AGE $ TREAT $ SCORES @@;
datalines;
HYB 17.5 HMA 22 HOC 23
NYC 23 NMB 14.5 NOA 14
LYA 12 LMC 12.5 L OB 5.5
;
PROC GLM;
Class IQ AGE TREAT;
MODEL SCORES = IQ AGE TREAT;
Means Treat/LSD;
Means Treat/Tukey;
RUN;
    
```

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